

Cloud Security Agent (Agent 10)

Deep Technical Specification & Operational Manual

Executive Overview & Purpose

The Cloud Security Agent performs deterministic auditing of infrastructure-as-code (Terraform, CloudFormation), container configurations (Docker), and cloud service posture to detect misconfigurations, exposed secrets, and privilege escalations across AWS, Azure, and GCP.

1. Architectural Overview & Workflow

- Position in LangGraph Engine: Second node in Pipeline A, executing after DevSecOpsAgent.
- Input Consumption: Consumes code audit findings (including secret-redacted file snippets) and raw IaC payloads.
- State Propagation: Writes `state.cloud_security_report` with per-service misconfiguration findings.

2. IaC Misconfiguration Detection

- Terraform/CloudFormation: Detects public S3 buckets, overly permissive security groups (0.0.0.0/0), unencrypted volumes, and IAM wildcard privileges.
- Docker Security Posture: Flags root execution, unversioned base images, missing healthchecks, and sensitive env passthrough.
- Service Posture: Checks AWS/GCP/Azure managed-service settings embedded in configuration code.

3. Deterministic Risk Assignment

- CIS Benchmark Alignment: Findings reference aligned CIS benchmark controls.
- Severity Classification: CRITICAL/HIGH/MEDIUM/LOW based on exploitability and blast radius.
- Remediation Guidance: Every finding ships with a concrete configuration fix snippet.